

HAHNENKAMM EHRENBACHHOEHE, Austria

WMO: 111350

Lat: 47.418N

Lon: 12.359E

Elev: 5786

StdP: 11.87

Time Zone: 1.00 (EUC)

Period: 94-10

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	4.9	7.9	-7.0	4.7	23.2	-2.5	6.0	21.9	22.7	24.4	18.9	25.5	3.9	270	0.676

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.1	68.1	56.1	65.7	55.3	63.4	54.0	58.5	65.1	56.7	63.1	55.3	61.8	5.4	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
56.0	83.1	61.5	54.0	76.9	59.6	52.3	72.3	58.3	28.1	65.5	26.8	63.3	25.8	61.9	70.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.8	15.1	12.8	1.4	72.5	5.1	3.9	-2.3	75.4	-5.3	77.7	-8.1	79.9	-11.8	82.8
			0.9	61.1	4.8	2.6	-2.6	63.0	-5.4	64.5	-8.1	66.0	-11.5	67.9
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	38.5	26.1	24.9	28.3	35.1	44.0	48.9	52.2	51.7	46.3	42.3	32.9	28.3
	DBStd	13.15	8.77	9.52	9.57	8.68	8.26	8.42	7.52	6.98	8.06	8.96	9.99	9.45
	HDD50	4651	742	704	672	451	222	121	66	65	159	260	516	673
	HDD65	9680	1207	1124	1137	898	651	484	400	413	560	705	963	1138
	CDD50	450	0	0	0	3	36	88	135	118	48	20	2	0
	CDD65	4	0	0	0	0	0	0	3	1	0	0	0	0
	CDH74	6	0	0	0	0	0	0	6	0	0	0	0	0
	CDH80	1	0	0	0	0	0	0	1	0	0	0	0	0
	WSAvg	5.6	6.0	6.0	6.0	5.8	5.7	5.5	5.6	5.3	4.9	5.3	6.0	5.6
	PrecAvg	51.5	2.6	2.4	3.3	3.3	5.3	6.6	7.1	6.9	4.8	3.7	3.0	2.6
(15) Precipitation	PrecMax	59.5	7.2	7.1	8.4	6.6	7.8	10.0	11.0	10.3	9.2	7.2	6.0	6.3
	PrecMin	42.4	0.3	0.2	1.1	0.2	1.3	3.8	2.2	4.3	2.2	0.9	0.0	0.3
	PrecStd	4.3	1.6	1.5	1.7	1.6	1.8	1.6	2.0	1.7	1.5	1.5	1.5	1.4
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	49.4	49.5	50.1	56.2	66.1	69.5	73.3	69.7	64.8	62.1	56.3	49.7
		MCWB	35.3	36.7	38.5	44.0	52.1	55.9	57.8	57.3	52.7	48.7	43.2	36.3
	2%	DB	44.8	45.0	46.6	52.8	61.9	66.5	69.1	66.4	61.7	58.4	52.3	45.8
		MCWB	32.7	33.0	37.1	42.1	50.3	55.2	57.5	55.7	51.0	46.7	41.1	35.4
	5%	DB	41.3	41.3	44.1	49.5	59.0	63.9	66.2	63.9	59.2	55.9	48.9	43.1
		MCWB	32.4	31.5	35.4	40.3	48.7	53.9	56.3	55.0	50.3	45.6	38.9	33.1
	10%	DB	38.2	37.9	41.5	46.7	55.7	61.2	63.3	61.5	56.9	53.5	46.4	40.5
		MCWB	30.7	30.6	34.1	38.5	47.0	52.2	55.0	53.9	49.2	44.9	37.4	31.9
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	37.7	38.0	40.5	45.7	54.1	59.2	62.2	59.4	55.6	51.2	45.4	39.3
		MCDB	44.8	46.8	46.3	53.5	62.6	66.2	67.4	65.2	60.3	56.9	54.1	46.6
	2%	WB	35.1	34.7	38.7	43.0	51.7	56.6	59.5	57.5	53.5	49.2	42.2	36.5
		MCDB	42.1	42.0	44.1	51.9	59.4	63.5	66.2	63.4	58.5	55.0	49.6	42.3
	5%	WB	32.9	33.1	37.0	41.1	49.5	55.0	57.3	56.1	51.8	47.6	40.1	34.9
		MCDB	39.3	39.4	42.0	48.0	56.7	61.9	64.0	62.0	57.2	53.5	46.5	40.8
	10%	WB	31.2	31.2	35.2	39.3	47.8	53.4	55.6	54.6	50.2	45.9	38.4	33.2
		MCDB	37.1	36.8	40.0	45.6	54.5	59.8	62.5	60.4	55.5	51.9	45.0	39.2
(35) Mean Daily Temperature Range	5% DB	MDBR	8.5	8.7	8.8	8.6	9.4	9.4	10.1	8.7	8.3	8.9	8.4	8.3
		MCDBR	12.1	10.9	10.6	10.1	11.8	11.9	13.5	11.0	10.4	10.5	9.8	10.4
		MCWBR	7.0	6.7	6.2	6.3	7.2	7.6	8.3	7.6	7.6	6.9	5.6	6.6
	5% WB	MCDBR	11.3	10.6	10.2	9.8	10.6	11.2	12.2	10.2	9.8	9.4	9.8	9.9
		MCWBR	7.4	7.5	6.3	6.5	7.4	7.9	8.7	7.8	7.6	6.7	6.4	6.9
(40) Clear-Sky Solar Irradiance	taub	0.235	0.255	0.287	0.316	0.328	0.334	0.334	0.330	0.305	0.283	0.259	0.235	(40)
	taud	2.458	2.409	2.404	2.424	2.439	2.448	2.439	2.471	2.528	2.572	2.568	2.506	(41)
	Ebn at Noon	281	295	299	298	297	294	292	290	289	282	270	269	(42)
	Edh at Noon	20	26	31	34	35	35	34	32	27	22	18	17	(43)
(44) All-Sky Solar Radiation	RadAvg	414	711	1025	1308	1489	1650	1673	1489	1091	763	430	337	(44)
	RadStd	50	89	120	166	178	172	162	176	144	81	70	53	(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (84 neighbors)	+0.76	N/A	N/A	N/A	+0.65	+0.65	N/A	N/A	+112	+52

Nomenclature: See separate page